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Risk shifting and disorganization in multi-tier contracting chains: The implications for public safety

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ABSTRACT

Organizational responses to financial crises have contributed to workplace changes, including a reliance on multi-tier sub-contracting chains. An extensive research literature discusses the adverse impact of these trends on worker health and safety. Research has shown that economic pressures leading to unsafe practices by contractors, hazardous forms of disorganization in contracting chains and regulatory failure contribute to adverse health and safety outcomes for workers. Less research has addressed the impact of sub-contracting chains on the potential for complex catastrophic incidents in which the public could be impacted. We contribute to a growing body of work on this subject by examining excavation work around high pressure natural gas pipelines in Australia. We draw on past research complemented with concepts from the economics literature to highlight the impact of 'risk shifting' as contractors seek to cope with financial pressure in highly competitive sectors. Contract terms incentivize contractors to work as quickly as possible while also prioritizing compliance with safety-related rules. Using qualitative data from 36 face-to-face interviews we discuss the way that supply chains characterized by these types of contracts can lead to situations where the pressure to work quickly means that contractors might trade off safe work practices in order to meet project deadlines and make a profit. We argue that in times of acute financial pressure, safety can become fungible with an associated increase in risk to the public. The findings of this research may have broader implications for other public risk situations in sectors with similar structures and organizational arrangements.

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1. Introduction

Organizational responses to events such as the global financial crisis and more localized economic fluctuations have contributed to key changes to employment relations. Over time, these changes have seen a growth in job insecurity and precarious employment with strategies previously considered 'non-standard' such as casual contracts, outsourcing and complicated sub-contracting chains now normal forms of employment relationships (Benach et al., 2000, 2007; Bryan and Rafferty, 2015; Johnstone et al., 2001). Notwithstanding the financial benefits these may provide to organizations, research shows that a reliance on sub-contracting strategies has important implications for workplace health and safety (WHS) (see Quinlan and Bohle, 2015, for a summary).

When it comes to large-scale organizational accidents, inquiries into several disasters have also highlighted problems with contractor management and communication. For example, the official commission of inquiry into the 2001 explosion at the AZF chemical factor in Toulouse, France that killed 30 people identified problems with contractor safety management as a critical factor in the incident. The inquiry further recommended that multi-tiered sub-contractor arrangements on major hazards sites be banned (Quinlan et al., 2013). Contracting and outsourcing is also ubiquitous in other hazardous industries, such as offshore oil and gas exploration (see also Wright, 1986, 1994). For example, drilling rigs are owned and operated by organizations that are then hired by the oil company that holds an exploration license in order to drill a series of wells to search for hydrocarbons. Other technically complex tasks such as well cementing and testing are undertaken by yet another group of specialist contractors. Many of the individual workers in this sector do not have stable employment and rotate through a series of employment relationships depending on broad economic and geographic factors. Blowouts in this industry including Deepwater Horizon (Hopkins, 2012) and Montara (Hayes, 2012) have raised questions of how risk is managed across such fluid contracting arrangements.

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Somewhat surprisingly given this background, the impact of outsourcing and sub-contracting on major accidents has been relatively little studied. Quinlan and Bohle (2008) report a total of 16 studies relevant to this issue with few seeking to explore the link between outsourcing and a drop in organizational safety performance (see also Quinlan et al., 2013). This paper aims to contribute another case study to this body of knowledge and to provide insights on the reasons for this link.

Our case study addresses the potential for damage to the high pressure natural gas pipelines that run under our cities, towns and suburbs (and across country) by those who regularly work around such infrastructure. Failure of a high pressure natural gas pipeline can have catastrophic consequences as was graphically seen at San Bruno California in 2010 when eight members of the public died and 40 homes were destroyed when a gas pipeline failed (Hayes and Hopkins, 2014). This particular failure was not the result of a pipeline strike, but such events are not uncommon. The potential consequences of gas pipeline failure were also dramatically illustrated in Ghislenghien, Belgium in 2004 when construction of a car park damaged a high pressure natural gas pipeline. When normal pipeline operations resumed some time later, the pipeline ruptured resulting in 24 deaths, mainly amongst emergency services crews (Mahgerefteh and Atti, 2006). From the perspective of the gas pipeline sector, non-pipeline contractors working near pipelines represent a risk to the integrity, and thus the safety, of pipelines, workers and the community (Ramírez-Camacho et al., 2017). Although external interference failure rates in Australia are low in comparison to other countries, this remains the dominant cause of pipeline damage, responsible for 85% of reported incidents (Tuft and Bonar, 2009; Tuft and Cunha, 2013).

One significant group of people who work around pipelines are civil construction contractors working for organizations with an interest in other types of linear infrastructure (electricity and communication cabling, water pipelines, roads, drains and the like). In this paper we draw on interviews with personnel from a case study organization (CSO) involved in a large-scale infrastructure project that includes work around buried assets, not limited to high pressure gas pipelines. The principle task for the CSO is rolling out their project on time and within budget and, typically, they use a competitive tendering process to contract out responsibility for excavation, construction and building work. As primarily project managers, the CSO operates a complex chain of Construction Partners (CPs) who engage and manage sub-contractors across the country. In the context of these complex relationships, we used qualitative face-to-face interviews to probe participants' awareness of the risks associated with working around hazardous buried infrastructure and the governance strategies they used to manage potential risks. Of particular interest was whether contracted time and cost pressures contribute to unsafe work practices in civil construction work around high pressure gas transmission pipelines.

To reflect on how work practices with the potential to impact safety outcomes are influenced by extended sub-contracting chains in the context of excavation near high pressure pipelines, we draw on two theoretical domains. First, the existing literature around worker health and safety in precarious and contingent work, including sub-contracting chains (e.g., see James et al., 2007; Quinlan et al., 2013; Walters et al., 2005, amongst others), which also has relevance to public safety. Complementing that approach, we use the concept of 'risk shifting' to investigate how financial and safety pressures are managed through the supply chain. The next section outlines the literature regarding worker health and safety outcomes in the context of sub-contracting chains and discusses the origin of the risk shifting concept and the way in which we apply it in this case.

2. Conceptual framework

Particularly in times of financial stress or crisis, extended sub-contractor supply chains provide a range of benefits for organizations, such as important cost savings, access to specialist skills, job creation, knowledge generation and innovation and enhancing the speed of task completion (Quinlan et al., 2013). These strategies represent a shift away from internal management hierarchies to a reliance on market forces (Williamson, 1975). The use of "fragmented organizational forms" (James et al., 2007, 165), such as sub-contracting, can be explained from either a "transaction cost" or "resource-based" perspective. From a transaction cost view, outsourcing or sub-contracting is one way to manage costs associated with production, whether internal or external, as well as service delivery. From a resource-based perspective, outsourcing and sub-contracting can offer ways to achieve better competitive outcomes based on the capacity of an organization to use resources (physical and non-physical) to secure a competitive advantage. In either case, the point here is that changes in the business environment contribute, amongst other things, to a transfer of risk to those suppliers (and workers) lower in the supply chain (James et al., 2007). Further, although providing benefits for client or contracting organizations this fragmentation does not come without potential repercussions for safety.

2.1. Outsourcing and worker safety

There is an extensive body of research into safety and contingent work that has identified contributory factors linked to work practices that are implicated in adverse health and safety outcomes for workers (Quinlan and Bohle, 2008; Quinlan et al., 2001). In their discussion of the impact of contingent work in the Australian long-haul trucking sector, Mayhew and Quinlan (2006) found that the associated competitive pressures had a direct negative impact on worker health and safety as well as exacerbated levels of risk. Quinlan et al. (2013) also discussed the adverse implications of outsourcing maintenance responsibilities in the context of serious aviation events, including accident fatalities, in the US aviation sector. As we noted earlier in this paper, issues around contractor safety management are also relevant for large-scale organizational accidents, with safety management in multi-tier contracting chains highlighted as critical factors (Quinlan et al., 2013, see also Wright, 1986, 1994). However, sub-contracting chains present particular challenges in terms of safety management and risk coordination. For example, control, and thus responsibility, is often diffused down the contracting chain and exacerbated by poor levels of supervision and training, which then prevents effective communication (Kochan et al., 1994; Rebitzer, 1995).

These issues, together with economic and financial pressures as well as other structural factors such as regulatory failure and the role of trade unions have been identified as having a significant impact on safety outcomes in sub-contracting chains (e.g., see Quinlan et al., 2001, 2013; Quinlan and Bohle, 2008). Similarly, Walters et al. (2005) note the intricacies of multi-employer work sites, with complex supply chains, large numbers of casual and relatively low skilled workers and low levels of union organization as significant challenges in terms of improving safety outcomes. The approach used in this paper does not discount the broad relevance of these factors and the implications for worker safety, but the focus here is specifically on work practices. The central point addressed in this paper is the way that economic and financial pressures can often lead contractors to engage in 'corner-cutting' or other unsafe work practices (James et al., 2007; Quinlan et al., 2013). Using the pipeline case, we draw on the existing literature

to expand the discussion beyond the implications for worker safety to argue that work practices driven by outsourcing are also relevant to public safety, at least in this case. Added to that, we use the interview data to expand on the literature and suggest that a particular "hazardous form of disorganization" (Quinlan et al., 2013, 285) emerges as an excessive focus on procedural compliance can contribute to a 'tick and flick' mentality. One unintended outcome is that such an approach detracts from safe work behavior particularly for workers dealing with tight project deadlines.

2.2. Risk shifting

As we have noted, changes to labor relations due to financial crises have seen organizations adopt a range of strategies, such as multi-tier contracting chains, in order to adapt to changing economic conditions. Another response identified in the economics literature is the way that financial crises, such as the 2007–08 global financial crisis (GFC), can exacerbate 'risk shifting' behavior in large financial institutions and financially distressed firms (e.g., see Eisdorfer, 2008; Gropp et al., 2011). In the financial sector, risk shifting refers to the way that organizations transfer, manage and mitigate known or foreseeable risks often through contractual mechanisms with the parties involved (e.g., see Shilliday et al., 2007). This concept of 'risk shifting' also provides a useful way to consider organizational and public safety implications that may arise in the context of our case study (for a discussion of risk shifting in other sectors, see Brennan and Lee, 2011; Dwyer and Phillips Lassus, 2015; Sidorov, 2011). Risk shifting and its consequences have been described as a process that occurs as governments "reconfigure their role in directions consonant with now dominant mantras of neo-liberal policy" (Gray, 2009, 326). While this paper does not attempt to critique neo-liberal economics and the development of WHS legislation (e.g., see Hutter, 1997, 2001; Tucker, 2003), the relevant point is that assumptions underpinning this view have real consequences for worker, and as we shall show, public safety outcomes.

In his discussion of a ticketing system applied to Canadian workers, Gray (2009) notes that a key concept under a neo-liberal framework is the notion that workers and employers are 'partners' in health and safety. He also points out that one outcome has been the 'responsibilization' of health and safety. Responsibilization means that despite presenting employers and workers as 'partners' risk is shifted to workers who are increasingly responsible for taking adequate and appropriate actions to ensure their own safety. This situation sits in contrast to earlier periods when employment relationships were characterized by what has been described as 'risk-sharing' in areas such as income security, retirement, sickness and health care (Bryan and Rafferty, 2015). Broader political and economic factors have seen these types of initiatives largely 'cashed out,' with risks once carried by the employer, and the state, increasingly shifted to workers (Benach et al., 2007; Bryan and Rafferty, 2015). This is important because workers, or in this case sub-contractors down the supply chain, often lack the necessary skills or resources to manage these risks (e.g., see Garland, 1996).

More specifically for this paper, risk shifting or the responsibilization of risk ignores workplace power relationships, including in extended sub-contracting chains with multiple stakeholders. This is important because ignoring power relationships means that the subtle 'coercions' of the workplace that 'silence' complaints, which places workers at greater risk, are not adequately considered (Gray, 2009). For example, in their discussion of the shipping industry Bhattacharya and Tang (2012) noted adverse outcomes for workers when managers ignored seafarers' concerns and demanded compliance. More than that, risk shifting or responsibilization strategies hide the fact that unsafe acts have a variety of organizational, or structural, causes such as production pressures due to the economic climate and that these can create work environments conducive to unsafe work practices (Economic Development Committee, 2004; Gray, 2009; Silbey, 2009). In a highly competitive contracting environment, there is the potential that the unequal power relationships inherent in these interactions can skew the structure or content of these mechanisms to the disadvantage of those lower down the supply chain. This point is well examined in the literature, which notes that not only does the 'power disparity' mean that costs and risk are transferred down the supply chain, but that this can lead to work intensification and precarious employment (Bhattacharya and Tang, 2012). This is important in terms of worker health and safety outcomes, as the employment conditions of contingent workers may mean that they decide not to voice their concerns (Aronsson, 1999). Thus, while financial crises may come and go the changes they bring to labor relationships, such as extended sub-contractor supply chains, remain. This includes the potential for adverse safety outcomes for organizations, workers and the public.

This is not a critique of the validity of risk shifting as a risk management strategy in all situations. Rather, the issue here is the way in which contracts that incentivize on time project delivery, which characterize multi-tier sub-contracting chains, 'shift' risk down the supply chain to workers that are ill-equipped to manage these risks and that this has implications for safety outcomes. An extensive body of research regarding supply chain and sub-contracting issues considers the negative outcomes for work health and safety as well as for organizational accidents. However, as Quinlan et al. (2013, 284) note, much of the existing work around safety in the context of extended sub-contractor chains tends to occur in 'industry silos,' which restricts the ability to learn from research and policy responses in diverse contexts. The case study discussed here, namely an organization carrying out a large-scale excavation project and the activities of civil construction contractors around high pressure natural gas transmission pipelines, contributes to the existing literature by illustrating a useful way in which to bridge the gap between organizational structure and worker experiences.

3. Methodology

To develop coherent understandings and explanations (Ackroyd and Hughes, 1992) regarding the impact of financial pressures on safety through the supply chain this research uses a qualitative framework consisting of in-depth semi-structured interviews. This research generally follows qualitative methods widely used in sociological research (Silverman, 2001), sensemaking research (Weick, 1995) and in safety research (Hopkins, 2006). Given the lack of research data in this area, use of an in-depth qualitative approach to data collection is appropriate to identify specific issues that could be studied further using quantitative methods and a larger sample of infrastructure projects and contract workers/organizations across a range of sectors.

The data presented here is drawn from a research project exploring attitudes to risk and prevention of encroachment on high pressure gas transmission pipelines by non-pipeline industry workers (McDermott and Hayes, 2014, 2016). Funding for this research was provided by the Energy Pipelines Cooperative Research Centre (EPCRC),[1] which is supported by the Australian Commonwealth government and in-kind support from the pipeline industry. As a result, the task for the major research project was to investigate issues that are of concern to industry partners. In this case, the priorities of the pipeline industry were to learn more about the conceptualizations of risk held by non-pipeline industry groups

[1] See http://epcrc.com.au.

that might come into contact with high pressure gas transmission pipelines. The primary research question that the major project sought to address was how non-pipeline industry groups (and their workers) comprehend risk and responsibility for the prevention of pipeline damage or failure. To answer this question, we sought to determine the motivations and tasks for any work activities around pipelines, the risk governance framework used not only to safeguard workers but also the public as well as any concerns or issues that emerged in daily work practices.

To operationalize the data collection we used open-ended semi-structured interviews. As demonstrated in other qualitative research (e.g., see Grogan et al., 2006), a semi-structured interview format enabled us to cover relevant interview topics while avoiding control over the interview or leading respondents in any particular direction. The interview data was segregated into four key topic areas, which were (1) Motivations for work (tasks) around pipeline installations; (2) Risk Governance (health and safety regulations); (3) Risk Governance and Supervision (how work is conducted and overseen) and (4) Worker Experiences and Concerns. Here, we discuss the interview data for the themes of Risk Governance in terms of health and safety and Worker Experiences and Concerns as identified by our interviewees. The level of analysis discussed in this paper focuses on the perceptions and experiences of workers as they go about their tasks as well as the choices they make in the social and organizational context within which they are situated. Approval for the research was obtained from the RMIT Human Research Ethics Committee and the recorded interviews were transcribed and then thematically coded for analysis. As part of this process, one variable that emerged from the interview data is the way that financial pressures and organizational imperatives to manage, or shift, risk in order to meet project deadlines had implications for safe work practices.

This paper discusses a small 'slice' of the interview data, namely the comments of case study organization (CSO) representatives, their Construction Partners (CPs), civil construction contractors and field workers involved in excavation work around buried infrastructure. A total of 36 interviews were conducted across these groups. Although the majority of interviews were conducted 'one-on-one' with the participant and the researcher, in some instances, interviews were conducted with a group of individuals and this resulted in a total of 38 interview participants. For the CSO, the data is drawn from 12 interviews with 13 representatives ranging from senior management through to field supervisors. Most CSO interviewees had been with the organization for no more than five years, reflecting its primary role in delivery of a major infrastructure program. All brought with them considerable experience either in the core business area of the CSO or from the WHS sector. For the CPs, we spoke with three representatives, including two senior managers and a site inspector. Although the CP interviewees had been employed by their respective organizations for some time their involvement with the CSO project was similar to that of the CSO representatives that we interviewed. Despite this, they also brought with them considerable experience from a range of projects that involved work around buried infrastructure in other sectors. Interviews for the CSO and CPs were conducted in 2014 and 2015 and took place either in their respective corporate offices or on site with field supervisors and were typically around one hour in duration.

For the civil construction contractors, the data is drawn from 21 interviews with 22 individuals including field workers. The civil construction contractors and field workers interviewed for this research have many years of experience across a range of contexts (e.g. local government councils and utility companies) and tasks (e.g. distribution piping, deep sewers, water mains, roads, drainage amongst others). Some of the civil construction contractors were engaged on the CSO project, while others can be better understood as small to medium enterprises that primarily provide services to local government or other infrastructure providers. Irrespective of the size of the civil construction firms from which representatives were interviewed, these groups have a long history of working around underground assets. Interviews with members of this group took place in 2015 and were most often conducted on site. These interviews ranged in duration between half an hour to an hour, sometimes longer, depending on the time that individuals had available to speak with the researcher.

The diversity of individuals from the CSO as well as civil construction contractors and field workers means that our participants have sound knowledge of processes and procedures, both up and down the contracting chain, and are thus well situated to provide a perspective on the impact of those strategies on their work practices. More specifically, the comments of the CSO representatives, civil construction contractors and field workers helps to explore whether these individuals understand the risks presented by high pressure transmission pipelines or whether structural factors, such as time and cost project pressures through the supply chain, influence worker behavior and safe work practices.

4. Contracts, payment and risk

Unlike most hazardous infrastructure such as chemical plants or oil refineries, high pressure natural gas transmission pipelines are located largely in areas that are accessible to the public and to people from other industry sectors who may be working nearby. As such, the pipeline industry puts a lot of effort into engineering and procedural controls, including heavy walled pipe buried to significant depth with a route marked by signs and regularly patrolled. The pipeline sector also makes information about pipelines available to those planning excavation works and runs awareness campaigns about how to work safely around pipelines. Despite these measures in place, ultimately work done around pipelines is mostly initiated and controlled by people from outside the pipeline sector such as the CSO and civil construction contractors.

Work undertaken by civil construction contractors discussed here is organized and carried out within a complex network of organizations. The CSO has several Construction Partners (CPs) each of whom employs contractors and sub-contractors to undertake project work at numerous locations around the country. Work is competitively tendered and the overarching document mediating relationships at each level is the contract itself. In the context of our analysis, the contract addresses two key issues – payment terms for work required and compliance with safety rules. The relevant rules are found in the Australian model *Work Health and Safety Act 2011* (WHSA) and subordinate documentation. Importantly for this paper, requirements include the process for organizing excavation activities such as locating existing buried services, using the Dial Before You Dig (DBYD)[2] service and potholing using non-destructive methods such as digging by hand if necessary prior to any mechanical excavation (Safe Work Australia, 2013). In terms of contract payment terms for work done, at least for those engaged on the CSO project, contractors are paid per meter of trenching or drilling and installation of materials. Although this reflects the need of the CSO to keep project costs predictable and directly linked to productivity, the difficulty is that this strategy makes some of the work conducted by contractors' invisible, in particular pre-planning activities such as locating buried infrastructure. As well as managing costs, the interview data from this research shows that contracts are an important way in which the CSO manages safety risk. For the CSO, risk governance is articulated in contracts in terms

[2] The one-call system in Australia, see http://www.1100.com.au/#.

of compliance with procedures that have a basis in the WHSA. Contractual terms and conditions mandate that CPs and any sub-contractors they engage align their policies with the CSO's overarching risk-based compliance framework. Added to that, CPs and sub-contractor compliance is strictly monitored and audited by the CSO.

In summary, the relationships between the organizations involved in this work are mediated by a series of contracts, each reflecting competitive market forces and particular business interests (Williamson, 1975). As they engage in excavation work around high pressure pipelines the activities of the CSO, their CPs and sub-contractors have the potential to significantly impact public safety. The relationships between these organizations are mediated through a series of contracts. The contracts serve two key functions, which are to stipulate both the payment terms and also the processes around compliance with the WHSA. These two functions are interlinked but problematically for safety, as we will see, are treated by the CSO as if they are independent.

4.1. The case study organization (CSO) and risk governance

The risk governance strategy adopted by the CSO regarding the potential for damage to underground assets owned by others is to insist on procedural compliance from their contractors and sub-contractors. The CSO places a high value on compliance with safety rules and procedures that specify mandatory processes for working near underground assets. One senior management executive described the relevant procedures as "*rules that cannot be bent*" and as *"hard and fast rules of what people must do*" [CSO 01]. For all the CSO representatives that we spoke with, compliance with these rules is integrally linked to managing the business risk presented by an asset strike as well as maintaining productivity. As one senior management CSO representative stated: "*You reduce incidents, you reduce damages, you reduce claims, you increase productivity, and you increase everything*" [CSO 04].

In this view of safety, adherence to process, accompanied by a concern to manage liability, drives a push towards a reliance on audits (Bourrier and Bieder, 2013). The data for this research shows that contractors are subject to surveillance and monitoring with a reliance on both desktop and field audits. One CP described their audit process, including recorded safety observations that were 'like check lists' as well as unannounced "*minor and major inspections … we have leadership walks that senior management will go out and perform …*" [CP 03]. In addition, sub-contractors to CPs are also audited by the CSO. As CP 03 explained, "*I guess, just for their [CSO] peace of mind if they head to a site they know that all their ducks are in a row and they've got all their paperwork and things like that. They do go out and they do check it, they will check things.*" The point of interest here is that the audit processes remain focused on the importance of ensuring workers' compliance with the necessary procedures, as defined by the CSO and CP. For example, CPs described 'cardinal' or 'golden rules' that sub-contractors were required to follow in order to "*do the right thing*" [CP 03], including WHSA requirements, such as Safe Work Method Statements (SWMS), that are "*with them on the job … every day, they sign in on it …*" [CP 01].

Underpinning these systems and processes are strongly held views that equate procedural compliance with the prevention of pipeline or asset strikes. As one CSO representative explained: "*… we just oversee it based on these guys following their own processes. And they have great ones and it's proven, like probably 98% of the time, had they adhered to that, their own process, it wouldn't have happened*" [CSO 10]. For this person, 98% of the responsibility for pipeline strikes seems to be allocated to workers and only 2% to problems with processes. In other words, a risk governance framework in the form of systems and processes is equated with safety. However, the responsibility for compliance is clearly shifted to sub-contractors down the supply chain. All the CSO representatives were of this view, as this indicative comment illustrates:

> *Yeah, well that sub-contractor would be responsible for it because it's their responsibility to get the DBYD plans and locate the assets … The principal contractor would wear obviously some of that responsibility if there were a death or anything like that … that's why everyone has to have their due diligence up to the client, but the main person would be that sub-contractor.* [CSO 12]

This comment emphasizes that in the view of CSO interviewees that we interviewed, responsibility and accountability ultimately rests with the sub-contractor at the bottom of the chain to follow and ensure compliance with rules that are fixed because they have their origin in regulation. In this way, the CSO devolves responsibility for safe work systems and processes to their CPs, or principal contractors, as well as sub-contractors down the supply chain. This represents a clear example of Gray's (2009) 'responsibilization' in which workers have an increased responsibility for taking action to ensure their own safety and in particular balancing safety requirements with other potentially conflicting goals.

4.2. The contractors' dilemma – risk shifting

As noted earlier in this paper, Shilliday et al. (2007) have emphasized the extent to which risk shifting can be enacted through contracts and there is evidence of this approach in the case study discussed here. Shifting responsibility for the management of risk, both in terms of safety and cost down the contracting chain creates a dilemma for contractors. The civil construction contractors and field workers that we have interviewed indicate that, at least in some cases, it is not possible to follow the imposed rules and meet other critical business objectives such as making a profit. This is because the uncertainty surrounding the cost of compliance with CSO systems is borne by the people at the bottom of the chain.

In this context, there are two key elements identified by the civil construction contractors and field workers in our sample group. First, civil construction contractors engaged on the CSO project emphasized that poor payment rates are problematic and create difficulties around meeting project objectives. One contractor described the implications for their business in interactions with a CP:

> *We approached [ORG] saying, "You're going to have to pay us extra for all of the potholing because we are just losing, and we're not drilling." We're spending all day, instead of doing it; we should be able to drill 100 m a day without a problem even with potholing. We were spending two or three days trying to get ready for one 60 m drill. Our company was losing money hand over fist. We asked them, "You've got to pay us" and they just weren't interested, not interested. We said, "We can't do our job properly if we don't do this potholing but we're losing money" and they weren't interested in paying us to do excess potholing.* [Civil CONST 02]

That at least some members of the CSO are aware of the commercial impact of this is evident, as one CSO representative told us: "*… we expect our contractors to be diligent. But with the rates that are paid … when we have a Construction Partner … the rates that they pass onto their contractors, they take a lot of the margin out … it's metres in the ground to them, and while they're stopped and potholing and all of that, they're not getting paid for that. So it's a commercial impact*" [CSO 13]. As far as this CSO interviewee understands it at least, contractors are required to follow procedures and yet any variations in work that they are required to do as a result of those procedures are not paid for by the CSO. This

highlights a second problem identified in the data, namely, the processes around compensating contractors for 'extra' work that might be required in the course of the project but that is not included in the original contract. Although typically addressed using 'contract variations' (Quinlan et al., 2013), this can be a contentious and difficult process.

At issue is not workers' lack of awareness of either the risks or the WHSA requirements for working around buried infrastructure because: *"they [workers] all know it's the most important part of the job, to pothole, to find services before we dig around them, to excavate all by hand around the service if necessary"* [CIVIL CONST O8]. Rather, the problem was, as this participant went on to state: *"We don't get compensated for that"* [CIVIL CONST 08]. In a highly competitive civil construction sector, the difficult process of claiming contract variations means that some contractors may choose to carry out the work at their own expense in order to meet the project deadline and maintain relationships (Hinton and Hamilton, 2015; Quinlan et al., 2013). There is evidence of this situation in the interview data, as one civil construction contractor working on the CSO project explained: *"We have only stuck in it because we want to be involved, it nearly sent us broke, put it that way"* [CIVIL CONST 02]. Fundamentally, contracts mediate power relationships and, in a highly competitive commercial environment, this can mean that those with less power, usually those lower down the supply chain, are faced with difficult decisions. This was a view articulated by most of the civil construction contracts and their workers that we interviewed. As one field worker stated, *"At the end of the day, they're ... the bosses ... and that's where it gets power hungry, at the end of the day they can say, "Yep. No. We're doing it this way ..." it's either abide by it or look for different job, look for a different contract, unfortunately and that's the same with everything really"* [WORK 10].

The data suggests that these strategies have implications for public safety too. For some contractors, the combination of difficult contractual processes and pressure to meet project deadlines may mean that the work is not done or is done poorly. As this field worker stated:

> *... that's [*poor payment rates*] where all the risks, that's were all the stupidity comes in, because they [*contractors*] can't afford to do it, so they ... take shortcuts, and then they end up hitting stuff, or missing stuff, or someone ends up getting hurt.* [WORK 07]

The civil construction contractors and field workers that we interviewed described their experiences with difficult contract variation processes and, while not admitting any impact on *their* ability to work safely, clearly indicated that this could be a problem for *other* contractors in the sector. As this contractor and a field worker indicated:

> *We spend weeks on some jobs, like when I say weeks I mean two, three weeks looking for assets that you can't find but we won't put the pipe in the ground. Whereas other contractors will spend two hours and go, "Stuff it, I'm putting it in" ... and if you're a contractor and you've got a business, you can't afford to have two weeks of downtime trying to find something that's not your issue.* [CIVIL CONST 11]

> *Like, we're thorough ... we don't cut any corners with safety ... it's not worth the risk. But other contractors out there are just doing all sorts ... without the plans, without the right gear ... just taking risks on things ... So obviously they're going to make money from cutting corners.* [WORK 08]

These comments reveal that financial pressures can result in less than optimal safety practices.

The concept of 'risk shifting' is a useful way to consider the way that adverse safety outcomes might result. The economics literature describes the way in which organizations use 'risk shifting' as a way to transfer risk and that this can be exacerbated in times of financial stress (e.g., see Eisdorfer, 2008; Gropp et al., 2011). However, the data presented in this paper shows that risk shifting behavior, legitimated by contractual mechanisms with specific safety requirements, can become a normalized part of labor relations. In terms of safety outcomes, the comments of our participants suggest that this process of 'risk shifting' also occurs as the costs of compliance with safety requirements are shifted down the chain and typically borne by the contractor.

4.3. Competitive tendering – impact on safety

So far we have seen that this case study provides another example in which the contractual relationships shift risk down the supply chain and place safety responsibility with those at the bottom. In addition, contractors face the dilemma of balancing contractually required but potentially time-consuming safety activities, enforced via compliance with rules, against financial incentives to work quickly. These mechanisms mean that workers are in a precarious employment situation and so may decide not to voice their concerns (Aronsson, 1999). A similar pattern emerges when we look at relationships at the tendering stage of work activities, suggesting that unequal power relationships exist right from the formative stage of organizational relationships.

A significant issue here is the focus on price. Of the civil construction contractors that we interviewed, all emphasized that being subject to project pressure also occurs during competitive tendering processes driven by cost, rather than safety, outcomes. For example, as these civil construction contractors stated:

> *It's all money-driven now. They say that when they assess a tender, price is only 30 or 40%. They have these breakdowns of how they evaluate the tender. Capability is normally 20%, quality is 20%, 15% is delivery and price is only 30%. But price is 100%.* [CIVIL CONST 08]

> *Anyone can ... put in a tender ... you can say you've got a bit of experience and at the end of the day price seems to be the main factor. If you're cheap enough you'll get the job because when they do tenders ... they've got a valuation process ... the price shouldn't govern whether they get the jobs, price, experience, you know, your workforce, who you've got, the equipment you've got ... They don't seem to care about those things. At the end of the day it's often who's the cheapest who'll get the job.* [CIVIL CONST 21]

Tendering processes also use a compliance based approach to safety matters. As reported to us, it is common to require extra, dedicated safety personnel in order to meet requirements for auditing amongst other elements. The cost of this must be included in the tender price which puts even more pressure on any allowance for extra safety-related planning activities that do not contribute directly to 'meters in the ground.' Several interviewees reported that increasing sums of money are spent on safety, but not in the right places and certainly not allowed for in a competitive tendering environment. This issue was not isolated to CSO contractors, as this non-CSO contractor and a CSO field worker stated:

> *... in the last five years, we've gone from having two safety officers within our organization to now to having four because of the compliance levels that must be undertaken ... it's five per cent of our turnover goes to compliance ... The customer's got to pay for this compliance levels that they demand and at the moment they're not paying for it.* [CIVIL CONST 20]

Safety is out of control. It's over the top. There's too many companies going broke lately because ... safety's over the top, and they don't get paid for it. ... They're losing money and then, because there's no work ... everyone's undercutting everyone, so someone tries to bill for that safety bloke in their quote but the other bloke doesn't, so they undercut them. They don't win the job ... Everyone's going broke because safety's gone out of control ... out here they expect the same standards and qualities and safety as [in] the mines, but they don't want to pay for it and that's why people are taking shortcuts, and stuff's getting hit. [WORK 07]

As the comment from CIVIL CONST 20 cited above suggests, for some civil construction contractors, there was a need for client organizations to accept some responsibility in their contracting strategies for a more equal distribution of the compliance costs necessary to ensure safe work practices.

For the civil construction contractors and field workers that we interviewed, a competitive tendering process that focused on price and that fails to adequately recognize the cost of compliance meant that the risk burden was shifted down the chain to contractors and field workers. The problem here was not the competitive nature of the sector but that contracting organizations prioritized savings over safety, which meant that some contractors traded off safety against other objectives. As this contractor stated: "... *[other contractors] are going in cheap, they're going in at a lower price so they need to take shortcuts to ensure that they do turn a profit in some shape or form ...*" [CIVIL CONST 20]. The civil contractors and field workers in our sample indicated their commitment to meeting their obligations in terms of WHSA procedures and requirements. However, as CIVIL CONST 20 cited above shows, there is a potential for unsafe work practices to occur as, in order to realize a financial return on the project, some contractors might be tempted to take safety short cuts.

The data shows that the CSO pushes responsibility for safety down the supply chain. Added to that, the conflict that exists between contract payment terms and safety compliance leaves contractors with the dilemma of balancing these risks in a way that is invisible and unreasonable. The question remains as to the effectiveness of such a way of organizing work. This brings us to hazardous forms of disorganization (Quinlan et al., 2013). We elaborate on that concept to consider the way in which a strict compliance-based safety focus, such as that evidenced by the CSO, constitutes a form of hazardous disorganization. This is because the 'real' factors that contribute to good safety outcomes are ignored as contractors and workers are pre-occupied with procedural compliance and completing the necessary paperwork so that they can continue with their work tasks.

5. Compliance is not enough

Quinlan et al. (2013) note the integral role of contracts in relation to sub-contracting arrangements. However, organizing work using these types of contracts fails to address the inherently conflicting requirements that are contained in these documents. In particular, payment arrangements that encourage contractors to work quickly mean that there is less time available to spend on safety-related planning and pre-work - for which they may not be paid. The case study discussed here demonstrates that financial pressures can lead to corner-cutting. Critically, in the case of hazardous infrastructure such as high pressure gas pipelines, the compliance-based approach adopted by the CSO transfers responsibility for risk to the public onto those lower in the chain. Managing the trade-off between public safety and the financial viability of their business has significant day-to-day implications for those at the bottom of the supply chain.

As we have seen, the CSO adopts a risk governance and management approach that requires strict compliance with rules[3] or procedures that are operationalised in contracts. Importantly, compliance is fundamentally linked with a concern to mitigate any business risk and as in important part of ensuring and maintaining productivity. Despite this confidence in the compliance-based model, the efficacy of such an approach is questionable. One CP stated that, "*okay we hit things ... thankfully we don't hit a lot of assets*" before going on to describe to us "*... the third hit in recent history*" [CP 01]. That this approach is not successful is also illustrated by a CSO manager who stated that, despite systems in place, that, "*we're still hitting things*" [CSO 01].

The safety research that we draw on here suggests that adherence to an overly compliance-based approach to safety does not achieve the best outcomes (e.g., see Bourrier and Bieder, 2013; Fucks and Dien, 2013; Hale and Borys, 2013; Hale and Swuste, 1998). This is problematic because viewing safety as a matter of compliance can result in a bureaucratic safety culture (Parker et al., 2006), with procedural compliance and correctly completed paperwork, or 'ticking the box,' prioritised above reducing risk using effective field practices. This is not to suggest that rules and procedures are not important. Rather, these play a key role in managing risk and are essential elements for any modern organization to function effectively (Bourrier and Bieder, 2013; Fucks and Dien, 2013). This is particularly the case for organizations that are responsible for ensuring that potentially hazardous infrastructure, such as pipelines, are kept safe. The point of note here is that rules, or procedures, can either act as an aid, supporting workers in a difficult task or as a standard that requires compliance, which can be enforced (Fucks and Dien, 2013). From this perspective, failure to comply or deviations from procedure are easier to understand. However, this also means that the social processes attached to the daily uncertainties and complexities, or the "real conditions under which safety is produced" are neglected (Bourrier and Bieder, 2013, 3). The literature shows us that in reality, safety results from the complementary interaction of formal prescriptions as well as informal rules (Fucks and Dien, 2013).

A reliance on procedures can create a context in which individuals forget, neglect or avoid anything that is not formalized. Increasing formalization of procedures, or bureaucratization, was identified in early organizational sociology literature, which pointed out that bureaucracy can lead to over-compliance. In that type of situation compliance with rules becomes the aim rather than a means to a goal (Merton, 1968). Such over-bureaucratization in an organization can result in the "transformation of people into robots" (Fucks and Dien, 2013, 32). This can contribute to a decline in actions that perform and enhance safety, such as the ability to consider a variety of elements, an analytical and critical sense at work including taking the initiative and questioning procedures that are seen as problematic, ineffective or too complex (Fucks and Dien, 2013). Added to that, and despite an intention to enhance safety, reliance on procedures and rules can contribute to a 'blame culture' with the potential for individuals to be stigmatized as responsible for any organizational failures (Reason, 1997). Reducing failure to the individual level in this way fails to consider broader organizational or social factors that might contribute to why individuals make 'wrong' decisions (Bourrier and Bieder, 2013; Fucks and Dien, 2013). There is evidence in the data that the focus of the CSO is on identifying individuals, or the 'human factors,' responsible for errors as the 'root

[3] For a discussion regarding safety rules see Hale, A., Borys, D., 2013. Working to rule, or working safely? Part 1: A state of the art review. *Safety Science,* 55, 207–221, Hale, A., Swuste, P., 1998. Safety rules: procedural freedom or action constraint? Ibid. 29, 163–177.

cause' of why accidents occur. As this CSO executive manager stated:

> *But you've got to remember . . . that these guys are labourers, right, and really what are they thinking? They're thinking I want to get in, get my days work done, pick up my pay at the end of the week and feed my family . . . I can go out to a building site now . . . and they can be excavating, and you'll still have guys in the trench with no helmet on. Now you and I can sit there and say, well, you know, Jesus, an excavator tooth through your head's not going to be pretty. But we still go out there and see that. Why? They know, they know, even when they see us coming, on goes the helmet. So, why, what drives that? . . . I mean I just want to get in and find out what are the human factors, why do people do this.* [CSO 01]

This comment demonstrates a process of risk shifting, or the responsibilization of risk, as the CSO allocates responsibility for accidents to the individual worker. From that perspective, descriptions of the 'wrong' behavior that contributed or lead to the error provide a basis for blame with correct behavior, i.e. following procedure the only way to eliminate error. Fundamentally, a safety strategy based on procedural compliance diverts safety responsibility away from management and on to workers (Fucks and Dien, 2013). That managing risk in this way is problematic in terms of safety is exacerbated in situations where workers are juggling competing demands of project deadlines and cost pressures.

A focus on compliance with procedure can also contribute to a form of hazardous disorganization. This is particularly the case as workers are more focused on completing the required paperwork in order to avoid blame for any errors or accidents rather than on completing their tasks safely. For some of the civil construction participants in our sample group, this was exacerbated by procedures perceived to be unnecessarily duplicated, ineffective or as creating project delays, which result in pressure to work quickly and thus potentially trade off safe work practices. As these participants indicated:

> *I guess it's their awareness and their focus gets shifted . . . your average maybe two or three man crew . . . they're blue-collar workers, they're used to excavating and doing their work. If their focus gets moved from what they know and they do well, now onto the paperwork, their focus gets lost on what they are doing and . . . they start to rush. They've still got to do the same amount of work in the same amount of time . . . the extra paperwork . . . if they're not reasonable really do make the guys lose focus.* [CIVIL CONST 13]

> *. . . they [processes] are a good thing. They do create a bit of a burden for the workers on site because it's all thrown at us and it all has to get done. But owners of companies, you're not allowed to tell them I said this, still expect the same amount of work to get done in the same time regardless of the extra paperwork and pressures involved . . . they get uppity and jump up and down at you because things haven't started when they think you should be on the drill rig drilling already.* [WORK 02]

The point here is not that procedures are not perceived as important or necessary elements linked to safe work practices. Rather, a perhaps unintended consequence is that ineffective, duplicated or unnecessary procedures can contribute to unsafe work practices. More problematically, a focus on procedures and paperwork carried the potential to create a 'tick and flick' compliance mentality. Rather than enhance safe work practices, this can be conceptualized as a form of hazardous disorganization as workers stop reading the required documentation, particularly when faced with tight project deadlines. For example:

> *"Here's a 42-page document, but just sign the back page." So guys just sign the back page* [CIVIL CONST 11]

> *. . . I guess for the newer guys, if you don't have that industry experience it can almost become very robotic, where they know where to go to tick and they know what's going to get them over the line. So it's almost robotic.* [CIVIL CONST 13]

These comments suggest that complex, or lengthy, paperwork carries the, unintended, potential to detract from safe work behavior. However, at the same time, reducing paperwork to an overly simplistic format creates a potential 'tick and flick' mentality as workers aim to 'get on with the job.' A 'tick and flick' mentality linked to an excessive focus on proceduralization appears to be another form disorganization along with those already highlighted by Quinlan et al. (2013) that is, weakened induction, fracturing of information flows and the inability of workers to organize themselves.

In summary, all the civil construction contractors and field workers that we interviewed acknowledged that safe work procedures were an important and necessary aspect of working safely around buried infrastructure, including high pressure gas transmission pipelines. However, to be effective, procedures needed to be appropriate to their work tasks and reflect the conditions encountered in their daily work activities. At issue for the civil construction contractors and field workers was the potential for unsafe work practices to occur as workers, frustrated by ineffective or overly complicated procedures, chose not to follow guidelines. In other words, an unintended consequence of a focus on procedural compliance is that these can detract from safe work behavior particularly for workers dealing with tight project deadlines. This can potentially lead to a 'tick and flick' mentality as workers are influenced by financial incentives and the importance of completing work tasks in order to meet project deadlines.

6. Conclusion

Labor relations are increasing characterized by strategies such as outsourcing and multi-tier contracting chains as organizations respond to economic fluctuations and financial crises (Benach et al., 2000, 2007; Bryan and Rafferty, 2015; Johnstone et al., 2001). This paper seeks to add to the body of knowledge regarding the impact of multi-tier sub-contracting chains on complex organizational accidents with the potential to impact the public. Building on the literature regarding the impact of outsourcing on worker safety (e.g. Quinlan et al., 2013), complemented by the concept of risk shifting from the economics literature (see Eisdorfer, 2008; Gropp et al., 2011), we considered the safety implications that may arise as a result of changing work practices as organizations shift the responsibility for managing risk down subcontractor chains in efforts to manage their own financial pressures. Importantly, decisions to adopt sub-contracting strategies do not mean that organizations can avoid or delegate responsibility for safety, either of individuals or organizational safety processes (Bryan and Rafferty, 2015).

To consider the impact of contracting strategies on risk shifting and the implications for safety we used qualitative interview data with representatives from a case study organization (CSO) and civil construction contractors involved in excavation work around high pressure natural gas pipelines in Australia. The implications of this case study are significant. Unlike many other hazardous industrial installations, high pressure natural gas transmission pipelines are located in public spaces and on private land where people from outside the pipeline sector live and work. In theory, pipeline organizations hold overall legal responsibility for safety of the public as

a result of their assets. Despite the engineering and procedural measures in place, in practice the pipeline sector does not have direct control over all activities near pipelines that pose risk to the public. Rather, the activities of non-pipeline groups working near pipelines are governed through contracts that, albeit unintentionally, push the responsibility for maintaining the balance between safety and cost to the lowest levels of the subcontractor chains. This is problematic, in part, because some of the essential tasks they must undertake to reduce risk are invisible due to the structure of their contracts.

The CSO discussed in this paper uses a risk governance and management approach that includes strict compliance with procedures that are operationalised in contracts that are awarded through a competitive tendering process. The 'risk shifting' concept provides a useful way to consider the implications of this approach for unsafe work practices. The civil contractors and workers in our sample stressed that they were committed to meeting their obligations under the WHSA. Of concern for these individuals was that responsibility for meeting compliance costs was distributed unequally, with the majority borne by contractors rather than client or contracting organizations. Exacerbating this compliance burden were competitive tendering processes where the cheapest price won out. This situation had the potential to create an 'inevitable' culture of non-compliance because some contractors would be unable to meet their safety compliance obligations for the cost of the successful tender and might be tempted to take safety short cuts.

Our study also provides insight into another mechanism that contributes to adverse safety outcomes in organizational relationships based on outsourcing. Many organizations including our CSO treat safety as a matter of compliance. This results in a bureaucratic culture (Parker et al., 2006) in which compliance with procedures and completing the associated paperwork are valued above actual field practices that may more effectively reduce risk. The data discussed in this paper suggests that a focus on compliance and completing paperwork, underpinned by workers concerns to avoid blame for errors, can be conceptualized as another form of 'disorganization' in multi-tiered contracting (Quinlan et al., 2013). The danger here is that a focus on procedural compliance, particularly in association with complex or lengthy paperwork, can detract from safe work behavior and contribute to a 'tick and flick' mentality, particularly for workers dealing with tight project deadlines.

This research broadens understanding of the extent to which workers are becoming increasingly responsible for the costs associated with their own safety and further, the safety of the public, due to their work. When it comes to the cost of accidents, research has shown that employers in Australia incur only 5% of the total costs incurred. Workers themselves bear a staggering 74% of the cost of accident and injury at work (Safe Work Australia, 2015). This study suggests that moves towards multi-tiered sub-contracting are moving the burden of preventative safety expenditure onto workers too. To give this further context, a recent CSIRO study on the future of work speaks glowingly of future work trends towards 'freelance workers' and notes that one in three workers in the US is now an independent contractor (Hajkowickz et al., 2016). A decade ago, high reliability researchers suggested that organizational interfaces (between departments, functions or organizations) are areas that pose a challenge for effective safety management (Bourrier, 2005). Such interfaces are multiplying in working arrangements such as we have uncovered in the civil contracting sector. This fragmentation of the economy poses direct challenges for prevention of rare but catastrophic events such as rupture of a high pressure gas pipeline.

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Appendix A

Examples of the semi-structured interview questions used to address the key research questions included:

Is the [organization/individual] aware of the potential for pipeline strike?
Is the [organization/individual] aware of the hazardous consequences of a serious pipeline strike?
What profile, if any, does this type of incident have in policy and work planning or management?
How is the potential for pipeline strike managed and by whom (e.g., management, corporate safety, engineering, etc.)?
How is responsibility for pipeline integrity addressed within the chain of contractors used to undertake field construction work?
What incentives do contractors have to consider safe options?
Do gas and oil pipelines feature in discussions about risk and safety?
What role, if any, do other stakeholders (e.g., local councils, regulators) play in [organization/individual] perceptions about pipeline risks?
What interfaces do third party activities have with the pipeline sector (in the field and in planning)? Are they effective?

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